The notation $\partial U/\partial \boldsymbol{\Theta}^k$ refers to a 3×3 matrix of partial derivatives of the potential with respect to the corresponding elements of the rotation matrix $\boldsymbol{\Theta}^k$, and, for a 3×3 matrix $\boldsymbol{A}$,

$$\mathrm{rot}(\boldsymbol{A}) = \mathrm{skew}^{-1}(\boldsymbol{A} - \boldsymbol{A}^T).$$

Then the algorithm proceeds by iteration of timesteps, as shown below.

DLM Rigid Body Molecular Dynamics Step

1. "Kick": update the momenta and angular momentum vectors with known forces and torques for a half step:

$$\boldsymbol{p}^k := \boldsymbol{p}^k + \frac{h}{2}\boldsymbol{F}^k, \qquad \boldsymbol{\pi}^k := \boldsymbol{\pi}^k + \frac{h}{2}\boldsymbol{\tau}^k.$$

2. "Drift": step forward a whole step in positions

$$\boldsymbol{q}^k := \boldsymbol{q}^k + h\boldsymbol{p}^k.$$

3. "Spin": update the rotational state of the kth rigid body by the sequence of steps

$$\begin{aligned}
&\boldsymbol{\Theta} := \boldsymbol{\Theta}^x\left(\tfrac{h}{2}\pi_1^k/I_1\right);\ \boldsymbol{\pi}^k := \boldsymbol{\Theta}\boldsymbol{\pi}^k;\ \boldsymbol{\Theta}^k := \boldsymbol{\Theta}^k\boldsymbol{\Theta}^T\\
&\boldsymbol{\Theta} := \boldsymbol{\Theta}^y\left(\tfrac{h}{2}\pi_2^k/I_2\right);\ \boldsymbol{\pi}^k := \boldsymbol{\Theta}\boldsymbol{\pi}^k;\ \boldsymbol{\Theta}^k := \boldsymbol{\Theta}^k\boldsymbol{\Theta}^T\\
&\boldsymbol{\Theta} := \boldsymbol{\Theta}^z\left(h\pi_3^k/I_3\right);\ \boldsymbol{\pi}^k := \boldsymbol{\Theta}\boldsymbol{\pi}^k;\ \boldsymbol{\Theta}^k := \boldsymbol{\Theta}^k\boldsymbol{\Theta}^T\\
&\boldsymbol{\Theta} := \boldsymbol{\Theta}^y\left(\tfrac{h}{2}\pi_2^k/I_2\right);\ \boldsymbol{\pi}^k := \boldsymbol{\Theta}\boldsymbol{\pi}^k;\ \boldsymbol{\Theta}^k := \boldsymbol{\Theta}^k\boldsymbol{\Theta}^T\\
&\boldsymbol{\Theta} := \boldsymbol{\Theta}^x\left(\tfrac{h}{2}\pi_1^k/I_1\right);\ \boldsymbol{\pi}^k := \boldsymbol{\Theta}\boldsymbol{\pi}^k;\ \boldsymbol{\Theta}^k := \boldsymbol{\Theta}^k\boldsymbol{\Theta}^T
\end{aligned}$$

   The notation $\boldsymbol{\Theta}^x(\phi)$ refers to a rotation around the x-axis by an angle ϕ, that is

$$\boldsymbol{\Theta}^x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\phi) & -\sin(\phi) \\ 0 & \sin(\phi) & \cos(\phi) \end{bmatrix},$$

   with corresponding formulae for $\boldsymbol{\Theta}^y$ and $\boldsymbol{\Theta}^z$.
4. "Kick": update the momenta and angular momentum vectors (as in Step 1) with a half step using new forces $\boldsymbol{F}^k$ and torques $\boldsymbol{\tau}^k$ evaluated at the current $\boldsymbol{q}$, $\boldsymbol{\Theta}$.

Note that the symmetric character of the algorithm could be revealed more clearly by dividing the "Drift" step into two half steps sandwiched around the Spin step (Kick, Drift, Spin, Drift, Kick), but as the updates Drift/Spin are decoupled, the order they are performed is not important.